



DO-IT-YOURSELF: PROJECT

Three red LEDs (LED1 through LED3) provide visual indication, driven by transistor T1. T1's base is connected to pin D4 of the Arduino Nano via resistor R1. Audio indication is provided by a 230V AC hooter, which is controlled by a 12V SPDT relay. Transistor T2 drives the relay, with its base connected to D11 pin of the Arduino Nano via resistor R3. The hooter connects to the normally-open (NO) contacts of the relay.

When the flame sensor detects a flame, it triggers two actions: the three LEDs light up for visual indication, and the hooter sounds for audio alert.

The source code for the project is written in the Arduino IDE platform. Before uploading the sketch to the Arduino Nano board, ensure the correct board and port are selected. Once confirmed, upload the sketch `Flamesensor.ino` to the board. Fig.

3 shows a screenshot of the source code.

Construction and testing

An actual-size, single-sided PCB layout for the fire alarm is shown in Fig. 4, along with its component layout in Fig. 5. After assembling the circuit on the PCB, it should be enclosed in a suitable box. The LEDs (LED1 through LED3) and CON2 should be fixed at the front of the box. The flame sensor should be connected across CON2 and positioned to ensure easy detection of flames.

Once the circuit is assembled, install the unit in an appropriate location where the flame sensor can effectively detect fire. To test the device, upload the source code to the Arduino Nano using the Arduino IDE. Afterward, disconnect the Nano board from the laptop or desktop, and connect 12V to the circuit.

EFY Note

The source code of this project is available for free download at www.electronicsforu.com

Bring a small flame (such as a lighter) near the flame sensor. The buzzer should sound, and the LEDs should light up, signalling flame detection. Once the flame is removed, the buzzer and LEDs should deactivate, confirming no flame is detected. The flame sensor is now operational and can be installed at a location as needed. **EFY**

Bonus. You can watch the video of the tutorial of this DIY project at <https://www.electronicsforu.com/videos-slideshows/make-your-own-frame-sensor>

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